

Kartik Sirohi

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RESEARCH INTERESTS

My research focuses on efficient inference and retrieval systems for large language models, spanning hardware-aware kernel optimization, LLM serving, and retrieval-augmented generation for multi-hop reasoning. I am particularly interested in closing the gap between algorithmic advances in deep learning and their practical deployment on real hardware. Alongside this, I am drawn to the mathematical foundations of learning systems: from formalizing utility and information value in retrieval to modeling learning dynamics in biological and artificial neural networks.

ACADEMICS

Bachelor of Technology in Information Technology

D.R. B.R. Ambedkar National Institute of Technology

2024-2028

Jalandhar, India

- **CGPA:** 9.47/10.00 (3.79/4.00) — Department Rank 1
- **Selected Coursework:** Discrete Mathematics (Graph and Group Theory), Design and Analysis of Algorithms, Probability and Computing, Operating Systems, Computer System Architecture.
- **Minor Project (Ongoing):** *Utility Inspired Retrieval System for Multi-Hop Reasoning.*

RESEARCH EXPERIENCE

Research Intern

QWorld

July 2026 – Present

Supervisor: Dr. Simranjit Singh

- **Project:** *QTagger+: Quantum Machine Learning for Ransomware Tagging and Classification with Generative Data Augmentation.*
- Developing the dataset curation pipeline for large-scale multi-family ransomware classification, including data collection, preprocessing, feature extraction, and quality validation.
- Collaborating with an international research team on the development of QTagger+, extending prior work on quantum malware classification through improved datasets and generative data augmentation.

AI Research Intern (Remote)

University of Galway, Ireland

January 2026 – March 2026

Supervisor: Dr. Priyanka Verma

- **Explainable Video Classification (ComSys 2026)** Contributed to the development of DICE-RoBERTa, an explainable video classification framework combining dense image captioning with RoBERTa-based action classification on the UCF101 benchmark accepted for presentation at **ComSys 2026**.
- **Model Development.** Implemented and evaluated components of the end-to-end PyTorch training and inference pipeline, including dataset preparation, experimentation, and model evaluation.

- **Experimental Evaluation.** Conducted extensive benchmarking and ablation experiments, analyzed per-class performance metrics, and participated in manuscript preparation for publication.
- **Computational Psychiatry.** Co-authored a second research project applying deep neural networks to computational psychiatry; manuscript currently under review at IEEE Access.

OPEN SOURCE CONTRIBUTIONS

ggml-org/llama.cpp — SIMD Kernel Optimization

PR #22209

Large Language Model Inference Engine

2026

- Implemented a SIMD-vectorized version of `ggml_vec_dot_q4_1_q8_1`, optimizing a performance-critical quantized dot-product kernel used during LLM inference.
- Worked through multiple maintainer review cycles before merge into the upstream `llama.cpp` codebase.

PyTorch

PR #172200

Deep Learning Framework

2026

- Identified and fixed a NaN propagation bug in the tensor `clamp()` implementation.
- Added regression tests and navigated PyTorch's large-scale review process to successful merge.

alexandria3k

PR #63

Scientific Data Engineering

2026

- Redesigned the lookup pipeline, reducing query complexity from $\mathcal{O}(n)$ to $\mathcal{O}(1)$.

SciPy

PR #23129

Scientific Computing

2025

- Added documentation examples for the Gauss–Jacobi quadrature API, improving accessibility of numerical integration utilities.

Tracer-Cloud/OpenSRE

PR #852, PR #917, PR #1059

Cloud Observability

2026

- Contributed unit tests and validation utilities for data processing pipelines and Grafana Tempo observability components.

SELECTED PROJECTS

tinyGPT: End-to-End Transformer Training and Inference Engine

GitHub

C++, Python, PyTorch, Transformer Architectures, Systems Programming

- Trained a decoder-only GPT language model (30M parameters) from scratch on the TinyStories dataset using PyTorch and developed tooling to export learned weights for native C++ inference.
- Built a high-performance C++ inference engine implementing multi-head attention, layer normalization, and autoregressive text generation.
- Implemented cache-efficient tiled matrix multiplication, SIMD-optimized kernels, and KV-cache-based incremental decoding to accelerate transformer inference on CPUs.

- Achieved **58.2 ms** Time-to-First-Token (TTFT) and **84.65 tokens/s** generation throughput on an Apple M1 CPU through compiler and systems-level optimizations.

mini-torch: Minimal Deep Learning Framework

[GitHub](#)

Python, Automatic Differentiation, Deep Learning

- Implemented a deep learning framework featuring reverse-mode automatic differentiation, a dynamic computation graph, tensor operations, and gradient accumulation from first principles.
- Developed SGD with momentum and Adam optimizers, a modular neural network training pipeline, and trained multilayer perceptrons to approximate continuous functions such as $\sin(x)$.

Athena: Local Retrieval-Augmented Search Agent

[GitHub](#)

Python, RAG, FAISS, Ollama, Sentence Transformers

- Developed a privacy-preserving retrieval-augmented search agent integrating web search, intelligent scraping, semantic retrieval, and local LLM inference.
- Implemented concurrent web scraping, dense embedding generation, FAISS-based semantic retrieval, and source-grounded response generation using locally hosted language models.

Linear Algebra Engine

[GitHub](#)

C++, pybind11, Numerical Computing

- Built a C++ linear algebra library with Python bindings supporting Gaussian elimination, Gauss-Jordan matrix inversion, tiled matrix multiplication, and determinant computation.
- Implemented cache-friendly blocked matrix multiplication and exposed the library to Python through pybind11 with a comprehensive automated test suite.

AWARDS & ACHIEVEMENTS

- **Academic Excellence Award** – Awarded by Dr. B. R. Ambedkar National Institute of Technology Jalandhar for securing the highest academic standing in the Department of Information Technology. *2025*
- **Third Place – HackMol 6.0** – Backend Engineer for *PETS*, an AI-powered pet adoption and rehoming platform. *2026*

TECHNICAL WRITING

- **Learning Representations by Back-propagating Errors** – An intuitive walkthrough of backpropagation and neural network training. [→ Read](#)
- **Graph Retrieval Methods: Retrieving Knowledge via Relationships** – An introduction to graph-based retrieval methods for Retrieval-Augmented Generation. [→ Read](#)
- **The Intuition Behind Neural Networks** – Visualizing parameter optimization during neural network training. [→ Read](#)